Are decomposition matrices of finite groups of Lie type unitriangular?

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Decomposition matrices

Let G be a finite group. Fix ℓ a prime and $(K, \mathcal{O}, \overline{\mathbb{F}_\ell})$ an ℓ-modular system. We let $V \in \mathrm{Irr}_K(G)$ be an irreducible KG-module and $W \in \mathrm{Irr}_{\overline{\mathbb{F}_\ell}}(G)$ an irreducible $\overline{\mathbb{F}_\ell}G$-module with P_W its projective indecomposable cover. The **ℓ-decomposition matrix** $D = (D_{V,W})_{V\in \mathrm{Irr}_K(G), W \in \mathrm{Irr}_{\overline{\mathbb{F}_\ell}}(G)}$ is given by: $P_W^{\mathcal{O}} \otimes_{\mathcal{O}} K = \oplus_{V \in \mathrm{Irr}_K(G)} D_{V,W} V$.

$$\begin{tikzcd} \mathcal{O} \arrow[r, hook] \arrow[d] & K = \mathrm{Frac}(\mathcal{O}) \\ \overline{\mathbb{F}_\ell} & \end{tikzcd}$$

An ℓ-modular system

$$\begin{tikzcd} P_W^{\mathcal{O}} \arrow[r] \arrow[d] & P_W^{\mathcal{O}} \otimes_{\mathcal{O}} K \\ P_W & \end{tikzcd}$$

PIMs to KG-modules

Finite groups of Lie type

For $\mathbf{G}$ a connected reductive algebraic group over $\mathbb{F}_q$ (q a power of a prime $p \neq \ell$, with p good for $\mathbf{G}$) and a Frobenius endomorphism $F : \mathbf{G} \to \mathbf{G}$, consider the **finite groups of Lie type** $\mathbf{G}^F$ and their interesting properties:

- Biggest class of groups in the classification of finite simple groups.
- Matrix groups such as $\mathrm{GL}_n(q)$, $\mathrm{Sp}_n(q)$, $G_2(q)$, $E_8(q)\ldots$ classified into 6 infinite classical families and 10 exceptional ones.
- Algebraic and geometric aspects/tools.

Unitriangularity?

Main Goal: Inductively deduce information on the $\overline{\mathbb{F}}_\ell G$-modules from the better known complex representation theory of G.

Evidence: True for symmetric groups (Brunat–Gramain–Jacon 2023).
True for $\mathrm{GL}_n(q)$ (Dipper 1985).
True for $\mathrm{GU}_n(q)$ (Geck 1991).

Conjecture (Geck 1990)

The ℓ-decomposition matrix of $\mathbf{G}^F$ is lower-unitriangular.

ℓ-Blocks

The decomposition matrix has a block decomposition.
$\longrightarrow$ Focus on one block (or a union of blocks) and in particular on the **unipotent ℓ-blocks** because:

- Number of PIMs in the unipotent ℓ-block does not depend on q.
- Almost all other blocks behave similarly to unipotent ℓ-blocks.

Strategy

Step 1. Find the number n of PIMs in the unipotent ℓ-blocks.

Step 2. Find candidates $V_1, \ldots V_n \in \mathrm{Irr}_K(\mathbf{G}^F)$.

Step 3. Find candidates for the projective modules $P_1, \ldots, P_n$.

Step 4. Check that $(\langle V_i, P_j^{\mathcal{O}} \otimes_{\mathcal{O}} K\rangle)_{1\le i,j\le n}$ is lower-unitriangular.

For the other blocks, a similar strategy should work.

Steps 1 and 2

We distinguish between two cases:

When ℓ is good, a basic set of candidates exists thanks to Geck–Hiss (1991, 1997). It consists of the **unipotent characters**.
They are parametrised thanks to Lusztig (1984).

When ℓ is bad, a basic set and its parametrisation has been conjectured by Chaneb (2020). It consists of some **unipotent characters and some isolated characters** in series indexed by ℓ-elements.
The number n is known thanks to Geck–Hiss.

In both cases, Step 1 can be computed using CHEVIE.

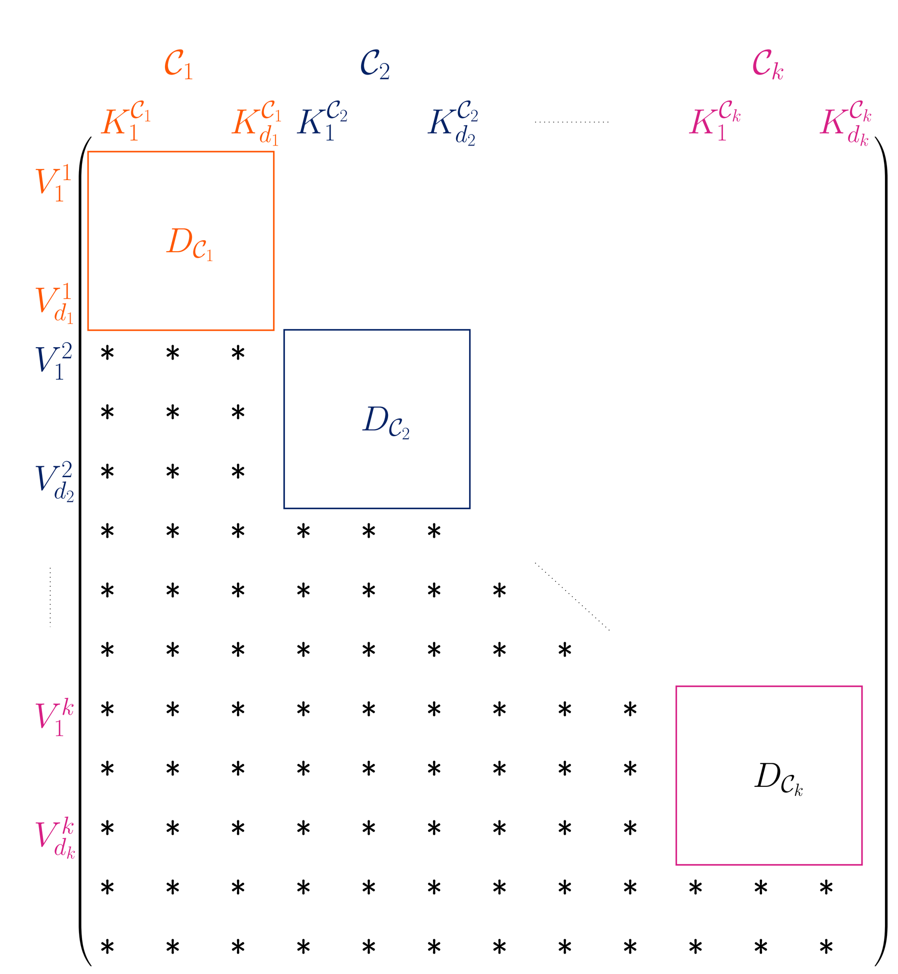
Step 3

A first class of candidates for the projective modules of $\mathbf{G}^F$ are the **Generalised Gelfand–Graev Characters** (GGGCs). Let $\mathcal{C}$ be a unipotent conjugacy class of $\mathbf{G}$ ($\mathcal{C} \in \mathrm{Uc}(\mathbf{G})$). If $\mathcal{C}$ is F-stable, then $\mathcal{C}^F$ decomposes into $n_{\mathcal{C}}$ unipotent conjugacy classes of $\mathbf{G}^F$.

For $1 \le i \le n_{\mathcal{C}}$, we construct a projective module $\Gamma_i^{\mathcal{C}}$ of $\mathbf{G}^F$. The wave-front set of $V \in \mathrm{Irr}_K(\mathbf{G}^F)$ is the unique F-stable $\mathcal{C} \in \mathrm{Uc}(\mathbf{G})$ such that:

- there is $1 \le i \le n_{\mathcal{C}}$ such that $\langle \Gamma_i^{\mathcal{C}}, V\rangle \neq 0$,
- for any $\mathcal{C}' \in \mathrm{Uc}(\mathbf{G})$ such that $\langle \Gamma_j^{\mathcal{C}'}, V\rangle \neq 0$ for some $1 \le j \le n_{\mathcal{C}'}$, we have $\dim(\mathcal{C}') \le \dim(\mathcal{C})$.

Consider the $\mathcal{C}_1, \ldots \mathcal{C}_k \in \mathrm{Uc}(\mathbf{G})$ which are wave-front sets of our candidates from Step 2 and take their associated GGGCs. There might be more V_i with wave-front set $\mathcal{C}$ than corresponding GGGCs.
$\longrightarrow$ Decompose the GGGC into **Kawanaka characters** $K_j^{\mathcal{C}_i}$.



Decomposition matrix of the unipotent ℓ-blocks

We order $\mathcal{C}_1, \ldots, \mathcal{C}_k$ such that $\dim(\mathcal{C}_i) \le \dim(\mathcal{C}_{i+1})$. We need to check that the matrices $D_{\mathcal{C}_i}$ are lower-unitriangular.

In some cases, there are enough GGGCs and the $D_{\mathcal{C}_i}$ are lower-unitriangular:

- for $\mathrm{GU}_n(q)$ (Geck 1991),
- for $\mathbf{G}$ simple adjoint of type B, C or D and $\ell = 2$ (Chaneb 2018).

Step 4

Directly computing the matrices $D_{\mathcal{C}}$ is difficult. We instead take the Fourier transform of Kawanaka characters and decompose them into characteristic functions of character sheaves (with unipotent support dual to $\mathcal{C}$).

When ℓ is good, we apply a general result of Lusztig (2015) for unipotent characters (sheaves).

When ℓ is bad, we study each case individually using a formula for the restriction of character sheaves.
This might allow us to treat the isolated blocks as well.

Theorem (Brunat–Dudas–Taylor 2020)

Assume that p is good and $p \neq \ell$. The ℓ-decomposition matrix of the unipotent ℓ-blocks of $\mathbf{G}^F$ is lower-unitriangular when ℓ is good and $\ell \nmid |Z(\mathbf{G})/Z(\mathbf{G})^\circ|$.

Theorem (R. 2024)

Assume that p is good and $p \neq \ell$. Let $\mathbf{G}$ be an adjoint simple group of exceptional type. The ℓ-decomposition matrix of the unipotent ℓ-blocks of $\mathbf{G}^F$ is lower-unitriangular.

References

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